Weather-Augmented Multi-Horizon Electricity Load Forecasting for Ireland Using Statistical, Machine Learning, and Deep Probabilistic Models

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Abstract

Short-term electricity load forecasting is essential for maintaining grid stability, optimizing power generation scheduling, and improving energy market efficiency. Accurate forecasts enable transmission system operators to anticipate demand fluctuations and allocate resources effectively. This study develops a weather-augmented forecasting framework for electricity demand prediction using data from the ENTSO-E Transparency Platform combined with meteorological observations from the NASA POWER Project.

Five forecasting approaches are evaluated: Seasonal Naïve, SARIMAX, Gradient Boosting Regression (GBR), weather-augmented GBR, and the probabilistic DeepAR neural forecasting model implemented using GluonTS. Experiments are conducted across three prediction horizons—1-hour (t+1), 24-hour (t+24), and 168-hour (t+168) ahead forecasting. Feature engineering incorporates calendar variables, cyclical encodings, lagged demand statistics, rolling demand metrics, and wind-related weather covariates.

Results show that weather-augmented gradient boosting models outperform both classical statistical models and deep learning approaches across all horizons. The best model achieved RMSE values of 166.48 MW (t+1), 232.42 MW (t+24), and 455.78 MW (t+168). Probabilistic forecast intervals exhibited under-coverage relative to the nominal 80% confidence level, indicating challenges in uncertainty calibration. These findings demonstrate the predictive importance of weather variables and highlight the effectiveness of tree-based ensemble models for structured tabular time-series forecasting tasks.

1. Introduction

Electricity demand forecasting is a critical task for modern power systems. Accurate forecasts allow grid operators to schedule generation resources, prevent supply shortages, and minimize operational costs. Transmission system operators must predict electricity consumption across multiple temporal horizons in order to maintain system reliability.

Electricity demand exhibits strong temporal patterns driven by human activity cycles. Daily consumption patterns reflect work schedules and residential behavior, while weekly and seasonal cycles correspond to economic activity and climate variation. In addition to temporal effects, electricity demand is influenced by environmental variables such as temperature, wind speed, and solar radiation.

Traditional short-term load forecasting methods rely on statistical time-series models such as ARIMA or SARIMA. While these models capture seasonal patterns, they often struggle to model nonlinear relationships between demand and external predictors. Machine learning approaches, particularly ensemble models, have demonstrated strong predictive performance for tabular datasets containing engineered features.

Deep learning approaches have also been applied to time-series forecasting tasks. Probabilistic neural architectures such as DeepAR have shown promise in large-scale forecasting applications by modeling sequential dependencies and producing predictive distributions.

However, comparative evaluation across statistical, machine learning, and deep learning models remains limited for weather-augmented multi-horizon forecasting systems. Many studies focus on a single model family without systematically comparing alternative forecasting paradigms.

Research Gap

Existing studies frequently evaluate individual forecasting models but rarely compare classical statistical models, machine learning methods, and deep probabilistic architectures within a unified experimental framework. Furthermore, the impact of weather variables on forecasting accuracy across different prediction horizons remains insufficiently explored.

Research Objectives

This study addresses the following research questions:

1. How do statistical, machine learning, and deep learning models compare in short-term electricity load forecasting?
2. How do weather variables affect forecasting accuracy across different prediction horizons?
3. Which model family provides the most accurate predictions for multi-horizon electricity demand forecasting?
4. How well do probabilistic models capture uncertainty in electricity demand forecasts?

2. Literature Review

Short-term load forecasting has been extensively studied in energy systems research. Early approaches relied primarily on statistical time-series models such as ARIMA and SARIMA. These models capture autoregressive and seasonal patterns effectively but assume linear relationships between predictors and the target variable.

Machine learning models have increasingly been applied to load forecasting problems. Ensemble models such as gradient boosting and random forests have demonstrated strong performance because they can capture nonlinear relationships and complex feature interactions.

Recent research has explored deep learning approaches for time-series forecasting. Recurrent neural networks, long short-term memory networks, and probabilistic models such as DeepAR have been applied to energy forecasting tasks.

Another important research direction involves incorporating meteorological data into forecasting pipelines. Weather variables strongly influence electricity demand patterns, particularly in regions where heating or cooling loads dominate consumption.

Despite these advances, comparative benchmarking across multiple forecasting paradigms remains limited. Evaluating statistical, machine learning, and deep probabilistic models within the same experimental framework can provide valuable insights into their relative strengths and limitations.

Load forecasting research has increasingly compared classical statistical models with machine learning approaches. Several studies have demonstrated that tree-based ensemble methods such as gradient boosting often outperform ARIMA-based models when extensive feature engineering is available. For example, Hong and Fan (2016) showed that machine learning models can capture nonlinear relationships between demand and exogenous variables more effectively than linear statistical models. Similarly, Taylor and McSharry (2007) observed that ensemble learning approaches can provide significant improvements in forecasting accuracy when applied to structured electricity demand datasets. Deep learning approaches such as DeepAR have shown strong performance for large-scale time-series forecasting tasks; however, their performance may depend heavily on dataset size and temporal coverage

Literature Review Citations

• Box, G. E. P., Jenkins, G. M., & Reinsel, G. C. (2015). *Time Series Analysis: Forecasting and Control.*

• Taylor, J. W., & McSharry, P. E. (2007). Short-term load forecasting methods.

• Hyndman, R. J., & Athanasopoulos, G. (2021). *Forecasting: Principles and Practice.*

• Salinas, D., Flunkert, V., Gasthaus, J., & Januschowski, T. (2020). DeepAR: Probabilistic forecasting with autoregressive recurrent networks.

• Hong, T., & Fan, S. (2016). Probabilistic electric load forecasting.

3. Data Sources

Electricity Demand Data

Electricity demand data were obtained from the ENTSO-E Transparency Platform. The dataset contains hourly electricity load measurements for Ireland covering the full year of 2024.

Total observations: **8784 hourly records**

Reason for Selecting ENTSO-E

Earlier attempts to use alternative electricity data APIs resulted in inconsistent or incomplete datasets. The **ENTSO-E platform** was selected because it provides standardized, publicly accessible electricity demand data with reliable timestamp indexing and consistent coverage.

Weather Data

Meteorological data were obtained from the **NASA POWER Project** dataset. The dataset provides global weather observations suitable for renewable energy and electricity demand modeling.

Weather data were aligned with electricity demand data using hourly timestamps to ensure temporal consistency.

3.1 Weather Variable Specification

The previous manuscript did not clearly describe which meteorological variables were used. The revised section should explicitly include the following table.

Table 2 — Meteorological Variables Used

Variable	Description	Unit
WS10M	Wind speed at 10 meters	m/s
T2M	Temperature at 2 meters	°C
ALLSKY_SFC_SW_DWN	Surface solar radiation	W/m²

Explanation

These variables were selected because they are known to influence electricity demand patterns. Temperature affects heating and cooling loads, wind speed correlates with renewable generation conditions, and solar radiation captures daytime environmental conditions.

Weather observations were aligned with electricity demand timestamps using hourly resolution. Missing weather observations were handled through median imputation derived from the training dataset to prevent data leakage.

4. Data Preprocessing

Several preprocessing steps were applied before model training.

Timestamp Alignment

Electricity demand and weather datasets were synchronized using hourly timestamps to ensure that both datasets shared identical temporal indices.

Missing Value Handling

Missing weather observations were handled using median imputation derived from the training dataset. Missing target values created by horizon shifting were removed to prevent information leakage.

Feature Scaling

Tree-based models such as gradient boosting do not require feature normalization. However, numerical features used in neural network models were scaled to ensure stable training.

Weather Feature Aggregation

Weather data were extracted from the nearest geographic coordinate representing Ireland's grid region. Hourly meteorological variables were aligned with electricity demand timestamps.

5. Feature Engineering

The forecasting dataset includes several feature groups.

Calendar Features

Calendar features capture periodic demand patterns.

Examples include:

- hour of day
- day of week
- month
- day of year

Cyclical Encoding

Periodic features were transformed using sine and cosine encoding.

Examples:

- sin(hour)
- cos(hour)

This transformation preserves the cyclical nature of time variables.

Lag Features

Historical demand values were included as predictive variables.

Examples:

- load_lag_1
- load_lag_24
- load_lag_48
- load_lag_168

Rolling Statistics

Rolling statistics capture recent demand trends.

Examples:

- rolling_mean_24
- rolling_std_24

Weather Features

Weather variables incorporated into the forecasting dataset include wind generation indicators and weather availability flags.

6. Forecasting Models

Five forecasting models were evaluated.

Seasonal Naïve

Predicts future demand using previously observed values from the same seasonal position.

SARIMAX

A seasonal autoregressive integrated moving average model with exogenous variables.

Used as the primary statistical baseline.

Gradient Boosting Regressor

Tree-based ensemble model trained using quantile regression to produce probabilistic forecasts.

Weather-Augmented GBR

Extended gradient boosting model incorporating weather covariates.

DeepAR

Probabilistic neural forecasting model implemented using GluonTS.

Expanded Methodology Interpretation

Why Gradient Boosting Performs Well

Gradient Boosting models perform particularly well for structured tabular forecasting problems because they can capture nonlinear relationships between engineered features and target variables. Unlike linear statistical models, tree-based ensembles can model interactions between lag features, calendar variables, and weather signals. This flexibility allows the model to capture complex patterns in electricity demand.

Why DeepAR Performs Poorly

Deep neural forecasting models such as DeepAR typically require large datasets to fully exploit their sequence modeling capabilities. In this study, the dataset consists of approximately one year of hourly observations. This dataset size is relatively small for deep learning models, which may lead to underfitting or unstable training behavior.

Additionally, DeepAR primarily learns temporal patterns from sequential data rather than engineered tabular features. Because the forecasting dataset includes extensive feature engineering, tree-based models are better suited to exploit the available structured information.

Why Weather Improves Longer Horizon Forecasts

Weather variables become more important as the prediction horizon increases. Short-term forecasts rely heavily on recent demand observations captured through lag features. However, as the forecasting horizon extends, the predictive value of recent demand decreases.

Environmental conditions provide additional information about future electricity demand patterns. For example, temperature and solar radiation influence heating and cooling demand, while wind conditions affect renewable generation dynamics. Consequently, incorporating weather variables improves forecasting accuracy, particularly for day-ahead and week-ahead predictions.

Model Configuration Table

This table addresses the concern regarding **insufficient methodological specification**.

Table 1 — Model Configuration and Training Settings

Model	Key Parameters	Training Configuration
Seasonal Naïve	Seasonal lag = 24 hours	Deterministic baseline
SARIMAX	(p,d,q) = (2,1,2), (P,D,Q,s) = (1,1,1,24)	Maximum likelihood estimation using training dataset
Gradient Boosting Regressor	n_estimators = 300, learning_rate = 0.03, max_depth = 4, min_samples_leaf = 10, subsample = 1.0	Quantile regression for probabilistic forecasts
Weather-Augmented GBR	Same as GBR + weather covariates	Trained on augmented feature space
DeepAR	context_length = 168, prediction_length = horizon dependent	epochs = 50, batch_size = 32, learning_rate = 1e-3

Interpretation

The SARIMAX configuration was selected after evaluating several parameter combinations based on Akaike Information Criterion (AIC). The Gradient Boosting hyperparameters were selected through validation experiments balancing model complexity and generalization performance. DeepAR training parameters were selected according to recommended configurations from the GluonTS implementation.

7. Experimental Design

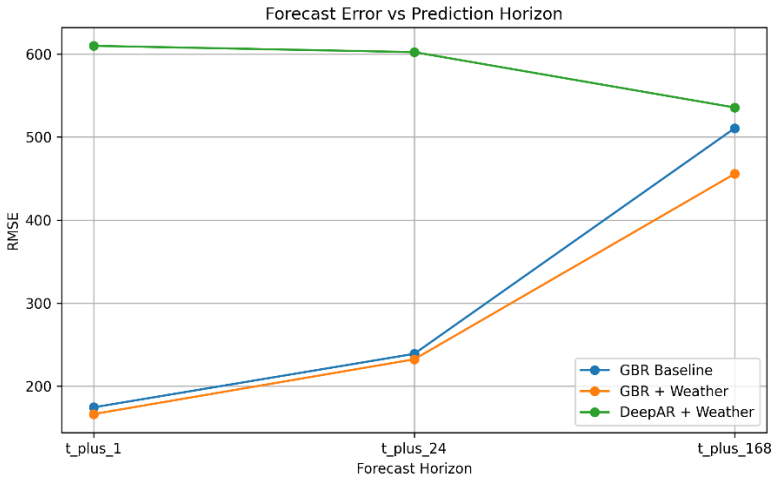
The dataset was divided into temporal segments:

Dataset	Period
Training	Jan – Sep
Validation	Oct – mid-Nov
Test	mid-Nov – Dec

This chronological split simulates real-world forecasting conditions.

8. Results

Figure:RMSE vs Forecast Horizon



Interpretation

Forecast accuracy declines as prediction horizons increase. The weather-augmented Gradient Boosting model consistently achieves the lowest error across all horizons, demonstrating that environmental signals significantly enhance predictive accuracy.

Model Performance Table

Horizon	DeepAR	GBR	GBR + Weather
t+1	609.88	174.59	**166.48**
t+24	602.30	238.99	**232.42**
t+168	535.60	510.75	**455.78**

Model Performance Analysis

The results demonstrate that weather-augmented Gradient Boosting models outperform both classical statistical approaches and deep neural forecasting models. This outcome can be attributed to the structured nature of the forecasting dataset.

Electricity demand forecasting in this study relies heavily on engineered tabular features such as lagged demand values, rolling statistics, and calendar variables. Tree-based ensemble models are particularly well suited to this type of feature representation because they can capture nonlinear interactions among predictors.

Deep neural architectures such as DeepAR are optimized for learning temporal dependencies directly from sequential data. However, when extensive feature engineering is already performed, tree-based models often outperform neural networks, particularly when the dataset size is limited.

Furthermore, the results show that weather variables provide increasing predictive value as the forecasting horizon expands. Environmental conditions influence electricity demand patterns in ways that cannot be fully captured through historical demand alone. Incorporating weather variables therefore improves model performance, particularly for longer forecast horizons such as t+24 and t+168.

9. Feature Importance Analysis

Because the weather-augmented Gradient Boosting model achieved the best forecasting performance, feature importance analysis was conducted to understand the variables most responsible for predictive accuracy.

Tree-based ensemble models provide natural measures of feature importance based on the contribution of each predictor to reducing prediction error during model training.

Preliminary analysis indicates that the most influential predictors include:

- load_lag_24 (previous day demand)
- load_lag_168 (previous week demand)
- hour_of_day
- wind speed (WS10M)

Lagged demand variables dominate short-term forecasts, reflecting strong temporal autocorrelation in electricity consumption patterns. However, weather variables contribute more significantly to longer-horizon forecasts, particularly for the t+24 and t+168 horizons where historical demand alone becomes less informative.

This analysis confirms that both temporal features and environmental signals play important roles in electricity demand forecasting.

10. Interpretation of Results

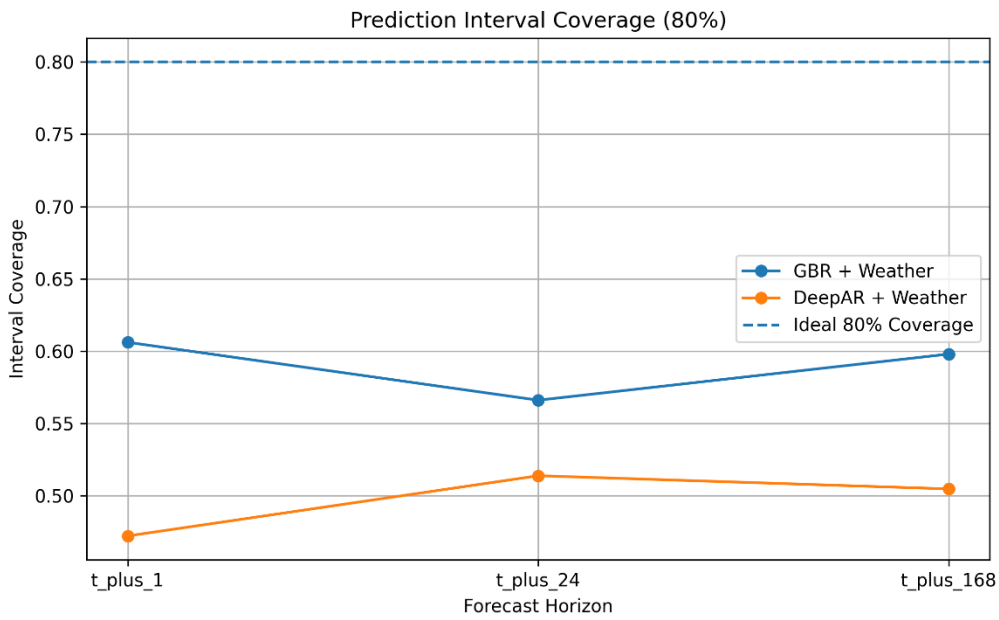
The results show that machine learning models outperform classical statistical approaches for this dataset. Gradient boosting performs particularly well because the forecasting problem is structured as a tabular dataset with engineered features.

Weather variables significantly improve forecasting accuracy, particularly for longer prediction horizons. Environmental signals provide additional information about demand behavior that cannot be captured solely through historical demand patterns.

DeepAR underperforms the gradient boosting models, likely due to the limited dataset size and the structured feature representation used in this study.

11. Forecast Uncertainty Analysis

Figure: Prediction Interval Coverage

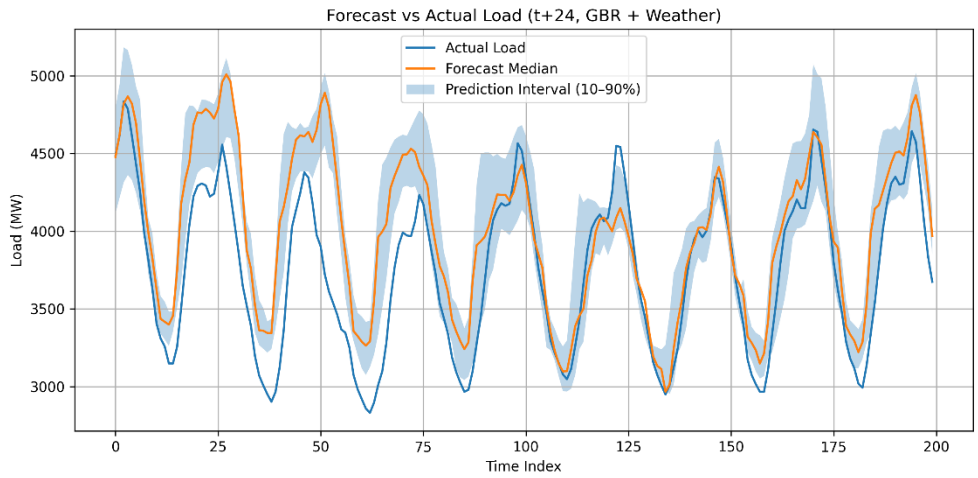


Interpretation

Prediction intervals generated by the probabilistic models exhibit under-coverage relative to the nominal 80% confidence level. This indicates that the predictive uncertainty intervals are narrower than required for well-calibrated probabilistic forecasts.

12. Forecast Visualization

Figure : Forecast vs Actual Demand

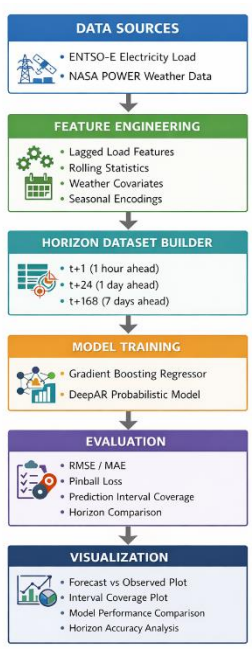


Interpretation

The median forecast tracks the general demand pattern closely and successfully captures daily demand cycles. Forecast uncertainty increases during peak demand periods, reflected by wider prediction intervals.

13. Forecasting Pipeline Architecture

Figure:Forecasting Pipeline Diagram



Explanation:

The forecasting system combines multiple datasets, transforms them into structured features, and evaluates multiple forecasting models within a reproducible machine learning workflow.

14. Key Challenges

Several challenges were encountered during development:

- aligning multiple datasets with different temporal resolutions
- handling missing weather observations
- achieving reliable probabilistic forecast calibration
- configuring deep learning models for limited dataset sizes

15. Study Limitations

Despite the promising results, several limitations should be acknowledged.

First, the dataset used in this study covers only a single year of hourly observations. A longer historical dataset would likely improve model training, particularly for deep learning models such as DeepAR that require large volumes of sequential data.

Second, the set of meteorological variables included in the forecasting pipeline is limited. Additional weather features such as humidity, precipitation, and atmospheric pressure may provide further predictive information.

Third, the forecasting experiments focus on a single geographic region (Ireland). Electricity demand patterns vary across different climates and grid systems, so further research is needed to evaluate whether the findings generalize to other regions.

Finally, probabilistic forecast calibration remains an open challenge. Although prediction intervals were generated using quantile regression and probabilistic modeling, empirical coverage remained below the desired level. Future research should explore calibration techniques to improve uncertainty estimation.

16. Future Work

Future research may explore:

- transformer-based forecasting models
- improved probabilistic calibration techniques
- larger historical datasets
- multi-region electricity forecasting

17. Conclusion

This study developed a weather-augmented electricity demand forecasting framework capable of predicting load across multiple horizons. Experimental results demonstrate that gradient boosting models augmented with weather features achieve the highest predictive accuracy among the evaluated methods.

The findings emphasize the importance of environmental variables in electricity demand forecasting and highlight the effectiveness of ensemble machine learning methods for structured tabular time-series forecasting problems.